

Playwright with TypeScript

Week 5 – Home Assignment



Program	Playwright with TypeScript
Module	TypeScript OOP Concepts
Week	Week 5
Assignment	Inheritance in TypeScript (Automation Testing Context)

1. Objective

To practice and understand inheritance in TypeScript by creating a parent class with common Playwright actions and child classes for specific page actions.

2. Problem Statement

Create a simple TypeScript program using inheritance to build a mini page action framework. Use a parent class for common actions and create child classes for Login and Dashboard pages with additional page-specific methods.

3. Requirements

- Create a parent class called **BasePage** with the following:
 - Property: `pageName` (string)
 - Methods: `openUrl(url: string)`, `getTitle(): string`, `click(locator: string)`, `fill(locator: string, value: string)`
- Create a child class called **LoginPage** that extends **BasePage** with the following:
 - Additional properties: `usernameLocator` (string), `passwordLocator` (string), `loginButtonLocator` (string)
 - Method: `login(username: string, password: string)` to simulate login using the parent class methods
- Create another child class called **DashboardPage** that extends **BasePage** with the following:
 - Additional property: `welcomeMessageLocator` (string)
 - Method: `verifyWelcomeMessage()` to simulate validation of the welcome message using parent class method `getTitle()`
- Create objects for both **LoginPage** and **DashboardPage** and call their methods to demonstrate inheritance.
- Add meaningful comments and use proper TypeScript types.

4. Sample Test Data and Expected Output

Sample Test Data (Code)	Expected Output (Console)
<pre>// LoginPage object const loginPage = new LoginPage("LoginPage"); loginPage.openUrl("https://openoscurce-demo.orangehrmlive.com/"); loginPage.login("Admin", "admin123"); // DashboardPage object const dashboardPage = new DashboardPage("DashboardPage"); dashboardPage.verifyWelcomeMessage();</pre>	<pre>LoginPage: Navigating to URL: https://openoscurce-demo.orangehrmlive.com/ LoginPage: Clicking on #username LoginPage: Entering text 'Admin' in #username LoginPage: Clicking on #password LoginPage: Entering text 'admin123' in #password LoginPage: Clicking on #loginBtn LoginPage: Login action completed for user: Admin DashboardPage: Validating welcome message DashboardPage: Page title: OrangeHRM - Dashboard DashboardPage: Welcome message is displayed successfully.</pre>

5. Technical Guidelines

- Use TypeScript (.ts file).
- Implement inheritance using the `extends` keyword.
- Create a parent class with common reusable methods.
- Implement child classes with page-specific methods.
- Use realistic method names and parameters related to automation testing (e.g., `openUrl`, `click`, `fill`, `login`, `getTitle`).
- Use `console.log()` to display the output (no need to use actual Playwright code).
- Add comments for better understanding.
- Follow proper naming conventions and code formatting.

6. Learning Outcome

After completing this assignment, you should be able to:

- Understand and implement inheritance in TypeScript.
- Create a parent class and child classes for real-time automation testing scenarios.
- Reuse parent class methods in child classes.
- Apply inheritance concepts in a page object style structure.